

Jules Berman

New York, NY. +1 917-209-8726

[julesmichaelberman@gmail.com] ▪ [website] ▪ [Google Scholar]

Summary

PhD candidate building probabilistic and generative models for data-driven and PDE-based physics simulation.

Author on **5 NeurIPS/ICML/ICLR papers** and **Harold Grad Prize winner** (top PhD '24). My work spans:

- **neural reduced order models that accelerate physics simulation** by $>100\times$ for parametric PDEs
- **time series modeling** — predictive approaches using CNNs, transformers, state-space models
- **models for stochastic simulation**—new generative approaches based on dynamic transport methods

Education

Ph.D., Computer Science — New York University, Courant Institute. GPA 3.9.

Expected May 2026

Awards: Harold Grad Memorial Prize (top PhD student, 2024) · MacCracken Fellowship (full award)

Advisor: Benjamin Peherstorfer

B.S., Computer Science — New York University.

May 2017

Awards: cum laude

Select Publications

Parametric model reduction of [...] stochastic systems via higher-order action matching

NeurIPS 2024

[arxiv] **J. Berman**, T. Blickhan, B. Peherstorfer.

CoLoRA: Continuous low-rank adaptation for reduced neural modeling [...]

ICML 2024

[arxiv] **J. Berman**, B. Peherstorfer.

Randomized Sparse Neural Galerkin Schemes with Deep Networks

spotlight ▪ **NeurIPS 2023**

[arxiv] **J. Berman**, B. Peherstorfer.

Stochastic Lifting for One-Step Per-Frame Video Generation and Physics Simulation

in review ▪ **ICLR 2026**

[arxiv] **J. Berman**, T. Blickhan, B. Peherstorfer.

Experience

Google

May 2025 – Aug 2025

Research Engineer Intern

Developed approach to generating time series embeddings via masked autoregression using large scale transformers; cut downstream model size 3-5 $\times$ while improving accuracy by 10-25% across datasets.

Meta

May 2024 – Aug 2024

Research Scientist Intern, CTRL-Labs

Designed and trained generative diffusion models of EMG data, improving downstream gesture classification accuracy by 10% through synthetic data augmentation.

Flatiron Institute

May 2021 – Aug 2022

Research Analyst, Center for Computational Neuroscience

Architected a distributed training platform for large-scale experiments (100k+ VAEs) and developed novel representational metrics enabling comparative analyses of latent representations.

Bloomberg LP

April 2018 – April 2021

Senior Software Engineer, Global Infrastructure Team

Deployed ML-based predictive systems to forecast future infrastructure demands from historical data.

Software: Python · JAX · PyTorch · Optax · Flax · NumPy · SLURM · Git · LaTeX

Projects: Developed an online free-to-play poker web application [justpoker.games]

Service: Section Leader Optimization and Computation Linear Algebra; Reviewer at NeurIPS, ICLR, ICML;